**AGA KHAN UNIVERSITY EXAMINATION BOARD**

**SECONDARY SCHOOL CERTIFICATE**

**CLASS X EXAMINATION**

**MAY 2011**

**Computer Science Paper I**

**Time allowed: 35 minutes Marks 25**

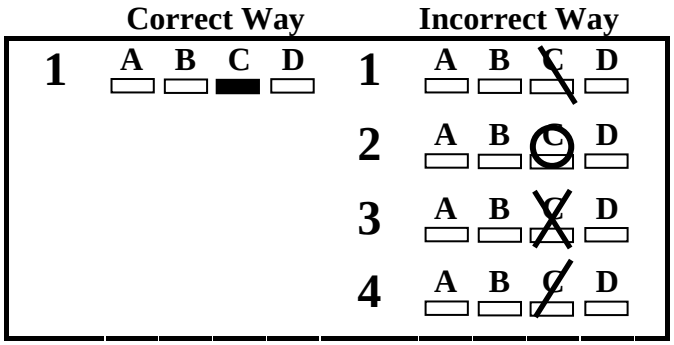
**INSTRUCTIONS**

1. Read each question carefully.

2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.

3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 25 only.

4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the box for your choice with a pencil as shown below.




5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new box.

6. DO NOT write anything in the answer grid. The computer only records what is in the boxes

1. Frequent use of computers may lead to all of the following problems EXCEPT

   A. blood pressure.
   B. eye-strain.
   C. backache.
   D. RSI (Repetitive Strain Injury).

2. Which of the following can detect the presence of a human being?

   A. Keypad
   B. Plotter
   C. Thermometer
   D. Motion sensor

3. When a hard copy is NOT needed, which of the following is used to get a picture?

   A. Plotter
   B. Monitor
   C. Scanner
   D. Printer

4. Internet contains a lot of information which can be viewed by everybody. If parents want to stop their children from accessing inappropriate websites, they can do this via

   A. a filter.
   B. a web browser.
   C. encryption software.
   D. a firewall.

5. Computer viruses are harmful to

   A. files and data.
   B. hardware.
   C. ALU.
   D. processor.

6. A step wise refinement of a set of statements is called a

   A. system.
   B. simulation.
   C. documentation.
   D. procedure.

7. In flow charts, a parallelogram represents either

   A. input or external storage.
   B. output or internal storage.
   C. input or output.
   D. input or loop.

8. Which of the following can have errors?

   A. Object code
   B. Compiler
   C. Source program
   D. Interpreter

9. Modifications in a program are made using

   A. a flowchart.
   B. user documentation.
   C. personal documentation.
   D. technical documentation.

10. Representation of the algebraic expression $\frac{2xy}{2x}$ in BASIC language is

    A. $(2xy)/(2x)$
    B. $(2*x*y)\backslash(2*x)$
    C. $(2*x*y)/(2*x)$
    D. $(2*xy)/(2*x)$

11. Consider a BASIC language expression $z = (x + y) + (x * y) + y$. If $x = 2$ and $y = 4$, then the value of $z$ is

    A. 20
    B. 18
    C. 16
    D. 0

12. Which of the following is an illegal variable name in BASIC language?

A. Total_cost
B. TOTALCOST
C. Totalcost
D. TotalCost

13. What is the output of the given program written in BASIC language?
10 Sales = 10000
20 If Sales> 10000 then Discount = Sales*20/100
30? Sales-Discount

A. 10000
B. 8000
C. 4000
D. 0

14. Which of the following is a conditional loop?

A. For..Next
B. While-Wend
C. Goto
D. If..Else..Endif

15. The Body Mass Index (BMI) categorizes a person according to the following table:

| BMI | Category |
|---|---|
| greater than 25 | over weight |
| between 25 and 19 (inclusive) | normal |
| less than 10 | under weight |

If a program is written in BASIC, which of the following statements is most likely to be used to display a category for a BMI?

A. For..Next
B. While-Wend
C. Goto
D. If..Else..Endif

16. Consider the following program:

For i = ? to? step?
Print i
Next i

What values will you assign to the loop to produce following series of numbers:
100, 98, 96, 94, ……. 6, 4, 2

A. For i = 2 to 100 Step 2
B. For i = 100 to 2 Step 2
C. For i = 2 to 100 Step -2
D. For i = 100 to 2 Step -2

17. Consider the following program:

10 For i = 0 to 1 Step .1
20 Print i
30 Next i

Which of the following output will appear?

A. 0,1
B. 0, .1, .2, .3, .4, .5, .6, .7, .8, .9
C. 0, .1, .2, .3, .4, .5, .6, .7, .8, .9, 1
D. 0

18. A one-dimensional array consisting of ten elements is declared as

A. X “10”
B. X (10)
C. X (10, 1)
D. X (1, 10)

19. Which statement of BASIC language will draw a diagonal line?

A. LINE (0,1) – (800,1)
B. LINE (0,2) – (800,2)
C. LINE (0,1) – (800,200)
D. LINE (0,1) – (400,1)

20. What will be the output of the following program?
10 country$ = “Pakistan”
20 l = len (country$)
30? l

A. Error
B. 7
C. 8
D. 9

21. Which of the following is the correct syntax to set the cursor to a particular location on the screen?

A. Locate 10,15: Print “Pakistan”
B. Locate 10;15: Print “Pakistan”
C. Location (10,15): Print “Pakistan”
D. Location 10,15: Print “Pakistan”

22. Which of the following is the correct syntax to extract three letters from the left hand side of the string?

A. Print left (“Pakistan”,3)
B. Print extract$(“Pakistan”,3)
C. Print left “Pakistan”,3
D. Print left$ (“Pakistan”,3)

23. Which of the following is used to take the cursor back to the previous location?

A. CALL
B. RETURN
C. GOSUB
D. JUMP

24. A Boolean expression using OR gate is

A. A + B
B. A*B
C. A/B
D. A.B

25. According to the rules of Boolean algebra, $A+\overline{A}$ is equal to

A. 0
B. A
C. 1
D. $\overline{A}$

END OF PAPER